Meeting Note 6

Date: 2026-01-29 **Time:** 10:00 - 11:00 **Location:** PQ703 **Attendees:** LIN Zhanzhi, CAO Yixin (Supervisor)

1. Objectives

- Review the results of the "Constant Tuning" phase.
- Finalize the values for `constants.hpp` before the large-scale benchmark.
- Discuss the impact of specific constants on recursive overhead and merge behavior.

2. Progress Report

- **Constant Tuning Completed**:
  - **`MIN_TRY_MERGE_SIZE`**: Optimized to **286**. Experiments showed this reduced runtime by ~13% (575ms -> 515ms for 10M random ints) by skipping the O(N) merge check for smaller ranges.
  - **`MIN_FIRST_RUN_SIZE`**: Verified at **16**. Larger initial runs didn't yield significant speedups for random data, but 16 was safe for structured data.
  - **`MIN_FIRST_RUNS_FACTOR`**: Updated to **6**. Shifted the crossover point for short runs to ranges of 64-128 elements.
  - **`MIN_PARALLEL_MERGE_PARTS_SIZE`**: Retained at **4096**. Smaller values (512) increased overhead without performance gain.
  - **`MAX_MIXED_INSERTION_SORT_SIZE`**: Set to **60**.
- **Code Cleanup**: Removed dead constants (`QUICKSORT_THRESHOLD`, `MAX_RUN_COUNT`) to clean up the codebase.

3. Discussion & Feedback

Key Discussion Points

- **Hardware Operation Costs**: Reviewed the finding that Comparisons are ~15-20x more expensive than Assignments (2.9ns vs 0.18ns) on the test hardware. This justifies the tuning decisions that slightly increase swap/assignment counts to save comparisons.
- **Micro vs Macro Benchmarks**: Confirmed that while micro-benchmarks (tuning specific constants) are good, the final validation must be a "Whole Benchmark" across ALL data patterns and sizes to ensure no regressions occurred in edge cases (e.g., Sawtooth, Organ Pipe).

Supervisor Feedback

> "The significant improvement (13%) from `MIN_TRY_MERGE_SIZE` is excellent. It demonstrates the high cost of the 'Check for Merge' logic in the original Java implementation when translated to C++." "Proceed with the final benchmark run immediately. Ensure you archive the tuning raw data as evidence for your final report."

4. Issues & Challenges

- **Benchmark Duration**: The full targeted benchmark (all types/patterns/sizes) is projected to take a very long time.

    - *Status: Resolved* - Reduced iterations from 30 to 10 for the large-scale run. This is statistically sufficient given the stability of the current measurements.

5. Action Items

- [ ] Execute the "Whole Benchmark" suite (Runtime & Operation Counts). (Due: 2026-01-30)
- [ ] Compile the final results into `summary_full.csv` and `metrics.csv`. (Due: 2026-01-30)
- [ ] Draft the "Implementation Section" of the Final Report based on these tuning findings. (Due: 2026-02-05)

6. Next Meeting Plan

- **Target Date:** 2026-02-05
- **Focus:** Review Final Benchmark Results and Draft Report.